



Experimentein.ai

Multimodal Similarity and Comparison of Protein Experiments

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Abstract

Protein-related experiments constitute the backbone of modern biological research, spanning expression studies, functional assays, binding experiments, and structural analyses. Despite their centrality, protein experiments remain scattered across unstructured scientific literature and heterogeneous datasets, making systematic discovery and comparison extremely difficult. *Experimentein.ai* proposes a multimodal experiment intelligence platform focused on protein experiments, enabling similarity-based search under acknowledged uncertainty, comparative analysis that surfaces missing metadata, and exploratory navigation that helps researchers understand what information is available versus assumed across previously conducted experiments.

1 Overview

Protein experiments represent a large proportion of experimental work in biology, including studies on enzymes, antibodies, structural proteins, and regulatory factors. These experiments are typically described in scientific papers through narrative text, figures, tables, and supplementary material rather than as structured, reusable records.

The motivation behind **Experimentein.ai** arises from three fundamental observations:

- Protein experiments are rarely exposed as first-class, searchable entities; instead, they are embedded in unstructured text and figures.
- Extracting protein experiments from papers is inherently difficult due to variability in experimental descriptions, protocols, and reporting standards.
- Even when similar protein experiments exist, researchers lack tools to systematically compare them across inputs, conditions, and observed outcomes while understanding what information is measured versus inferred.

As a result, experimental knowledge related to proteins is often rediscovered, duplicated, or underutilized.

2 Problem Statement

The problem addressed by **Experimentein.ai** spans multiple axes:

2.1 Experiment Fragmentation

Protein experiments are distributed across papers, datasets, and internal lab records. Each source uses different terminology, structure, and levels of detail, making unified access difficult.

2.2 Lack of Experiment-Level Indexing

Most search tools operate at the document level rather than the experiment level. Researchers searching for relevant protein experiments are forced to manually scan papers instead of querying experiments directly.

2.3 Difficulty of Experiment Comparison

Even when two protein experiments target the same protein or function, meaningful comparison is challenging due to differences in:

- Experimental conditions (temperature, pH, buffer, expression system)
- Protocol variations
- Measurement techniques
- Reported outcomes (quantitative vs qualitative)

Moreover, critical methodological information is frequently unreported in source materials. Researchers need to distinguish measured parameters from those that can only be probabilistically inferred, and understand the implications of missing data for experimental comparability.

2.4 Missing Interpretability and Provenance

Automated similarity methods often lack transparency. Researchers need to understand *why* two protein experiments are considered similar or different, which parameters are explicitly measured versus assumed, and where each piece of information originates. Additionally, they need visibility into missing metadata to assess the reliability of similarity judgments.

3 Alignment with Use Case 4

Experimentein.ai directly aligns with **Use Case 4: Multimodal Biological Design and Discovery Intelligence**.

The platform enables:

- Ingestion of protein-related experimental knowledge from heterogeneous sources, preserving both reported and unreported parameters.
- Multimodal similarity search across protein experiments using textual descriptions, experimental metadata, and associated artifacts, while explicitly flagging gaps and assumptions.
- Guided exploration of related protein experiments through similarity-based navigation that accounts for parameter certainty.
- Comparative analysis to highlight critical differences between experiments and surface missing metadata, supporting hypothesis generation and informed experimental planning.

Rather than predicting biological outcomes, the system supports discovery by organizing, contextualizing, and transparently representing existing protein experiments.

4 Use of Qdrant and Adaptive Data Representation

Qdrant is used as the core vector database to represent protein experiments as adaptive, sparse data objects rather than fixed dense embeddings. Each experiment is represented as:

- **Explicitly reported parameters:** Experimental text (methods, results, figure captions), protein-related entities and annotations, measured experimental conditions and outcomes.
- **Missing or unreported parameters:** Gaps are flagged explicitly rather than padded with defaults, preserving awareness of what information is unknown.
- **Probabilistic assumptions:** A biology-informed language model generates educated assumptions about unreported metadata, tagged with confidence scores and explicitly marked as inferred rather than measured.

This adaptive representation enables efficient similarity search, filtering, and neighborhood exploration while preserving the distinction between measured and assumed parameters. Similarity comparisons are calibrated to account for parameter certainty—comparisons are more confident when critical parameters are explicitly reported in both experiments, and are flagged as uncertain when assumptions are involved. Links to original experimental evidence are maintained throughout.

5 Experiment Ingestion Overview

Figure 1 illustrates a high-level view of how protein experiments are ingested from different sources into the system.

The ingestion pipeline extracts all explicitly reported experimental details into the adaptive data object. Unreported parameters are preserved as gaps rather than filled with defaults. Following extraction, the biology-informed assumption module evaluates what information can be probabilistically inferred from context (e.g., implicit standards in the field, author’s typical practices, publication date conventions, lab reputation patterns). These inferences are stored separately from measured data and tagged with confidence scores, allowing downstream systems and users to distinguish measured from assumed parameters.

6 Web Application Overview

Figure 2 presents a simplified view of how researchers interact with the platform.

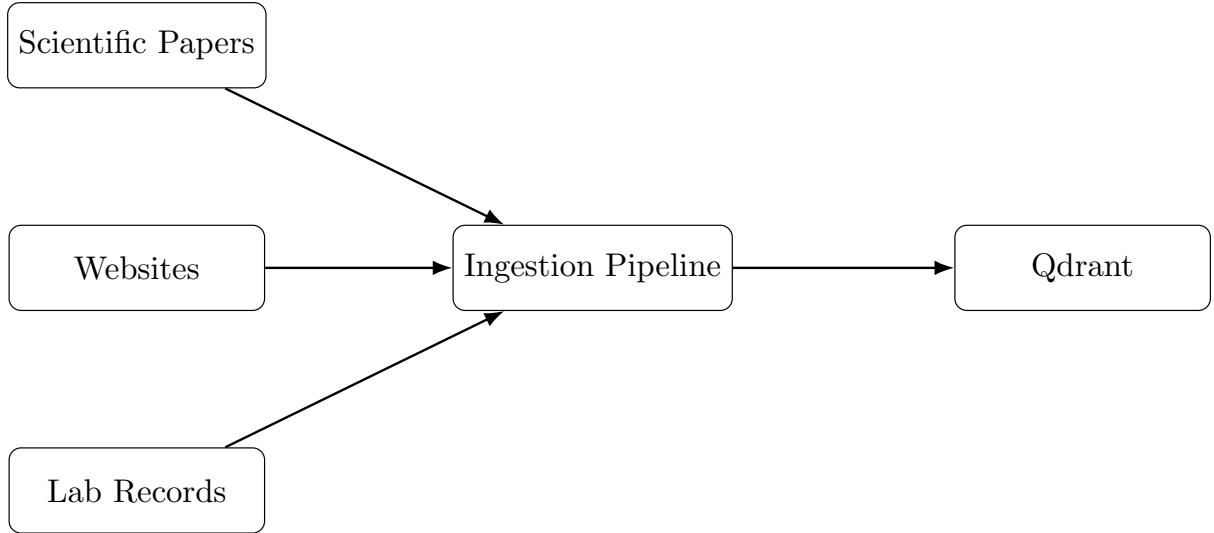


Figure 1: High-level ingestion of protein experiments from heterogeneous sources into Qdrant, with explicit preservation of missing metadata and probabilistic assumption generation.

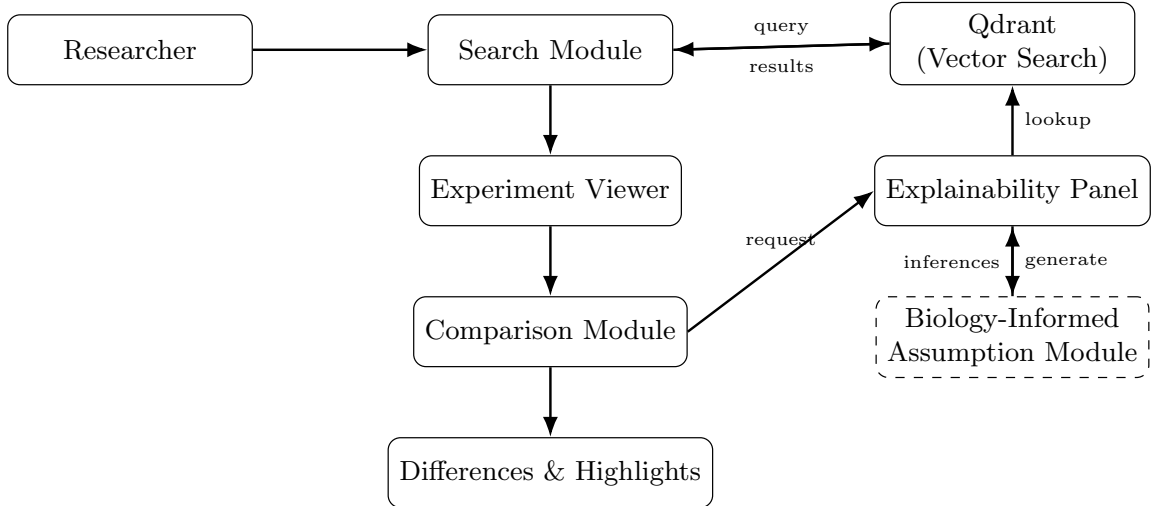


Figure 2: Web application flow of *Experimentein.ai*, showing search, visualization, comparison, and explainability with Qdrant and the biology-informed assumption module.

The Explainability Panel surfaces not just why experiments are similar, but which parameters are measured versus assumed, what metadata gaps exist in each experiment, and the confidence scores associated with any inferred parameters. This allows researchers to make informed judgments about whether similarity comparisons are reliable given the parameter certainty.

7 Disclaimer and Limitations

Experimentein.ai is designed as an experimental research prototype and does not aim to function as a biological inference or decision-making system. The platform does not claim to predict biological outcomes, establish causal relationships, or replace domain expertise or experimental validation.

Several limitations must be acknowledged:

- **No Biological Inference Guarantee.** The system focuses on organizing, retrieving,

and comparing existing protein experiments. Similarity-based retrieval and comparison should not be interpreted as biological inference, mechanistic explanation, or predictive modeling.

- **Explicit Handling of Incompleteness.** The system preserves the reality that source materials (papers, databases) are incomplete. Rather than obscuring this incompleteness through imputation or defaults, it explicitly marks unreported parameters and surfaces probabilistic assumptions about missing values. Scientists operate under acknowledged uncertainty rather than false confidence. The validity of any similarity comparison depends on the completeness and accuracy of reported data in the source materials and the reliability of probabilistic assumptions.
- **Domain-Informed but Not Infallible.** The biology-informed assumption module is trained on available domain knowledge but should not be treated as definitive. Scientists are responsible for validating assumptions against their own experimental context and laboratory protocols before drawing conclusions. The development team does not claim comprehensive biological expertise; certain biological nuances, experimental subtleties, or domain-specific assumptions may be oversimplified or misrepresented.
- **Time and Resource Constraints.** Due to the limited time frame of the hackathon, the implemented system represents a Minimum Viable Product (MVP). It does not cover the full scope required for a production-ready biological knowledge system and prioritizes conceptual validation over completeness.

These limitations emphasize that **Experimentein.ai** should be viewed as a proof-of-concept for experiment organization and comparison under acknowledged uncertainty, rather than a definitive biological intelligence system.

8 Conclusion

Experimentein.ai addresses a critical gap in protein-focused biological research by transforming fragmented experimental knowledge into a searchable, comparable, and transparently interpretable experiment space. By representing experiments as adaptive, sparse data objects that distinguish measured from assumed parameters, the system enables researchers to discover related experiments, learn from past work, and make informed experimental decisions grounded in honest assessment of what is and isn't known.

References

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